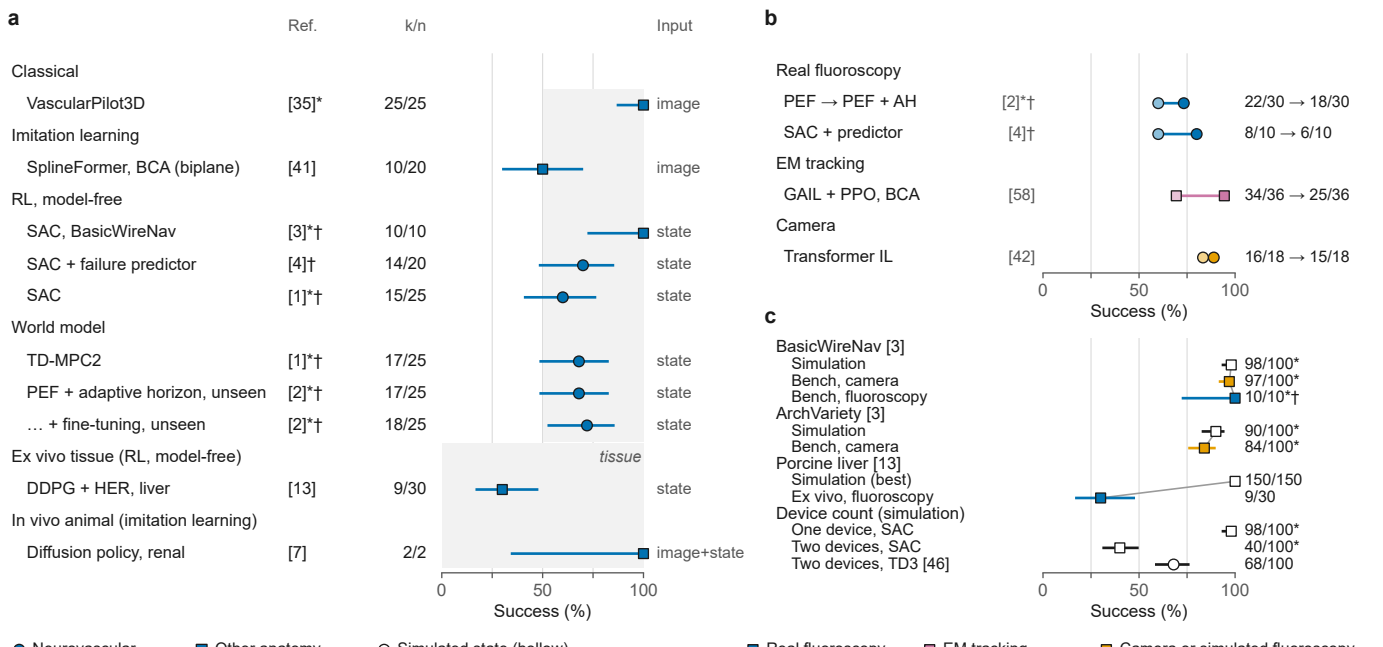

a
Ref.
k/n
Input
Classical
VascularPilot3D
[35]*
25/25
image
Imitation learning
SplineFormer, BCA (biplane)
[41]
10/20
image
RL, model-free
SAC, BasicWireNav
[3]*†
10/10
state
SAC + failure predictor
[4]†
14/20
state
SAC
[1]*†
15/25
state
World model
TD-MPC2
[1]*†
17/25
state
PEF + adaptive horizon, unseen
[2]*†
17/25
state
… + fine-tuning, unseen
[2]*†
18/25
state
Ex vivo tissue (RL, model-free)
tissue
DDPG + HER, liver
[13]
9/30
state
In vivo animal (imitation learning)
Diffusion policy, renal
[7]
2/2
image+state
0
50
100
Success (%)
b
Real fluoroscopy
PEF → PEF + AH
[2]*†
22/30 → 18/30
SAC + predictor
[4]†
8/10 → 6/10
EM tracking
GAIL + PPO, BCA
[58]
34/36 → 25/36
Camera
Transformer IL
[42]
16/18 → 15/18
0
50
100
Success (%)
c
BasicWireNav [3]
Simulation
98/100*
Bench, camera
97/100*
Bench, fluoroscopy
10/10*†
ArchVariety [3]
Simulation
90/100*
Bench, camera
84/100*
Porcine liver [13]
Simulation (best)
150/150
Ex vivo, fluoroscopy
9/30
Device count (simulation)
One device, SAC
98/100*
Two devices, SAC
40/100*
Two devices, TD3 [46]
68/100
0
50
100
Success (%)
Neurovascular
Other anatomy
Simulated state (hollow)
Bars: Wilson 95% intervals. * Pending field; † [5] tip tracker.
Real fluoroscopy
EM tracking
Camera or simulated fluoroscopy
(b) seen, solid fill; unseen, light fill